SERIES: A

QP CODE: 24900087

SERIES: A **Reg No:.....**

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS)

REGULAR EXAMINATION NOVEMBER 2024

First Semester

Multi-Disciplinary Course – MG1MDCECC100, MG1MDCECT104

CREATIVE ROBOTICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hours Maximum Marks: 35

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A
Multiple Choice Questions
Answer any 35 questions.
Each question carries 1 mark

1 Identify which component is the "brain" of the Arduino board. [U] [1]

a) Power supply b) Voltage regulator

c) Microcontroller d) USB port

2 Relate the purpose of the "Vin" pin on the Arduino board. [U] [1]

a) To supply 3.3V to external components b) To provide an external power source for the board

c) To interface with I2C devices d) To reset the microcontroller

3 Which port is used to upload code to the Arduino board? [K] [1]

a) Ethernet port b) USB port

c) Serial port d) HDMI port

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4 Identify the function of the RX pin on Arduino Uno. [U] [1]

a) To send digital data
b) To receive serial data
c) To read analog signals
d) To reset the microcontroller

5 Relate the Arduino IDE to its primary purpose. [U] [1]

a) To create websites
b) To compile, upload, and manage Arduino sketches
c) To monitor temperature sensors
d) To control external devices

6 Identify the correct syntax to set pin 13 as an output in Arduino. [U] [1]

a) pinMode(13, INPUT);
b) pinMode(13, OUTPUT);
c) pinMode(13, INPUT_PULLUP);
d) pinMode(13, ANALOG);

7 Identify the correct syntax to set pin 3 as an output in Arduino. [U] [1]

a) pinMode(3, INPUT);
b) pinMode(3, OUTPUT);
c) pinMode(3, INPUT_PULLUP);
d) pinMode(3, ANALOG);

8 Which tool is used to write and upload code to Arduino boards? [K] [1]

a) Python IDE
b) Arduino IDE
c) Code::Blocks
d) Visual Studio

9 If you connect an LED to pin 2 and use digitalWrite(2, LOW);, what would you expect to see if the LED was previously on? [U] [2]

a) The LED will blink
b) The LED will turn off
c) The LED will turn on
d) Nothing will happen

10 What will the following code do? PinMode(9, OUTPUT); digitalWrite(9, LOW); [U] [2]

a) Pin 9 outputs 5V
b) Pin 9 outputs 0V
c) Pin 9 will output 3.3V
d) Pin 9 will read input voltage

11 PWM in Arduino is used to simulate which type of signal? [U] [2]

a) LED ON and OFF
b) Pure analog signal
c) Pure digital signal
d) Analog signal using digital output

12 What is the range of values that can be written using the analogWrite() function? [K] [2]

a) 0 to 1000
b) 1 to 100
c) 0 to 10
d) 0 to 255

13 How would you calculate the output voltage of a voltage divider with resistances R1 = 10kΩ and R2 = 10kΩ, and input voltage = 12V? [U] [2]

a) 4V
b) 6V
c) 8V
d) 3V

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14 What voltage corresponds to a reading of 512 from analogRead() on a 5V Arduino? [U] [2]

a) 1.25V b) 2.5V

c) 5V d) 3.3V

15 How would increasing the baud rate from 9600 to 115200 affect the speed and reliability of data transmission in a noise-prone environment? [U] [2]

a) Speed increases but reliability decreases in noisy environments. b) Both speed and reliability improve.

c) The speed decreases, but the transmission becomes more stable. d) There is no impact on either speed or reliability.

16 Which of the following is the correct way to print text data on the serial monitor in Arduino? [U] [2]

a) Serial.print("Hello"); b) Serial.display("Hello");

c) Serial.show("Hello"); d) Serial.write("Hello");

17 Which statement correctly initializes a for loop in Arduino to run 10 times? [U] [1]

a) for (int i = 0; i <= 11; i++) b) for (int i = 1; i <= 9; i++)

c) for (int i = 0; i < 10; i++) d) for (int i = 1; i < 9; i++)

18 "In the following code, what is the value of i when the loop terminates? [U] [1]

```
for (int i = 0; i < 4; i++) {
  // Do something
}"
```

a) 3 b) 4

c) 5 d) 2

19 Which of the following is the correct syntax for a while loop in Arduino? [U] [1]

a) while i < 5 { // Do something } b) while (i < 5) { // Do something }

c) while (int i < 5) { // Do something } d) while (i > 5); { // Do something }

20 Which of the following correctly increments the loop variable i inside a while loop? [U] [1]

a) i + 1; b) i = i + 1;

c) i++; d) Both B and C

21 What will happen if you connect an LED directly between a digital pin and ground without a resistor? [U] [1]

a) The LED will work fine b) The LED will not light up

c) The LED may burn out d) The Arduino board will stop working

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22 How do you gradually increase the brightness of an LED using Arduino? [U] [1]

a) Use analogWrite() to adjust the duty cycle
b) Use digitalWrite() to set pin states repeatedly
c) Change the resistor value dynamically
d) Use pinMode() to adjust brightness

23 When does the code inside the setup() function run? [U] [1]

a) Every time the program loops
b) Only once when the Arduino board starts up or resets
c) Continuously until stopped
d) Before every loop iteration

24 How do you initialize multiple pins as outputs in a single line in Arduino? [U] [1]

a) pinMode(8, 9, OUTPUT);
b) pinMode(8, OUTPUT); pinMode(9, OUTPUT);
c) initialize(8, 9, OUTPUT);
d) multiPinMode(OUTPUT, 8, 9);

25 In Arduino, what default unit is the delay() function measured in? [K] [1]

a) Milliseconds
b) Seconds
c) Minutes
d) Microseconds

26 Which of the following is a correct syntax in Arduino programming? [K] [1]

a) delay(1000);
b) Delay(1000);
c) delay:1000;
d) delay;1000;

27 What is the role of 'int i = 0' in a FOR loop? [K] [1]

a) Declares and initializes a variable
b) Specifies the loop condition
c) Increments a variable
d) Declares a constant

28 Which of the following is true about a FOR loop in Arduino? [U] [1]

a) Executes a set number of times
b) Can run infinitely
c) Cannot be used with delay()
d) Runs only once

29 What type of waves does an ultrasonic sensor emit? [K] [2]

a) Radio waves
b) Ultrasonic waves
c) Electromagnetic waves
d) Sound waves

30 How is the distance calculated using the time taken by the sound wave? [K] [2]

a) Distance = Speed * Time
b) Distance = Time / Speed
c) Distance = Speed / Time
d) Distance = Time * Speed * 2

31 Which of the following can cause inaccurate readings in ultrasonic sensors? [K] [2]

a) Soft surfaces
b) Hard surfaces
c) Distance
d) Smooth surfaces

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32 What happens if pinMode(echoPin, OUTPUT) is used for the Echo pin? [K] [2]

a) It will correctly set the Echo pin
b) It will cause a syntax error
c) It will generate incorrect readings
d) Nothing will happen

33 Which pin on an MQ2 sensor outputs analog data? [U] [2]

a) D0
b) A0
c) GND
d) VCC

34 What precaution should be taken when using an IR flame sensor [U] [2]

a) Avoid using it with digital systems
b) Protect it from heat exposure
c) Do not use it near visible light
d) Expose it to sunlight for accurate readings

35 What does LDR stand for? [U] [2]

a) Light Delay Resistor
b) Light Developing Resistor
c) Light Driving Resistor
d) Light Dependent Resistor

36 What is the function of a relay in an Arduino-LDR circuit? [U] [2]

a) To measure resistance
b) To control high voltage devices
c) To decrease voltage
d) To lower resistance

37 What type of motor is used in precise control of angle? [U] [2]

a) DC Motor
b) AC Motor
c) Servo Motor
d) Stepper Motor

38 How many degrees of rotation can most standard servo motors achieve? [U] [2]

a) 0°-90°
b) 0°-180°
c) 0°-270°
d) 0°-360°

39 What does the L298N motor driver control? [U] [2]

a) Voltage
b) Current
c) Speed
d) Power

40 What is the correct method to stop a DC motor using an L293D driver? [U] [2]

a) digitalWrite(pin, LOW);
b) analogWrite(pin, LOW);
c) analogWrite(pin, OFF);
d) motorControl(pin, STOP);

(35 × 1 = 35)

END OF THE QUESTION PAPER

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	a) 3.3V	b) 2.5V		
	c) 1.25V	d) 5V		
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	a) Use pinMode() to adjust brightness	b) Change the resistor value dynamically		
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29	"In the following code, what is the value of i when the loop terminates?
	for (int i = 0; i < 4; i++) {
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			[U]	[1]

	a) 4	b) 2		
	c) 3	d) 5		
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	a) 3V	b) 6V		
	c) 4V	d) 8V		
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	a) Power supply	b) Voltage regulator		

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c) Microcontroller d) USB port

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a) pinMode(3, INPUT_PULLUP); b) pinMode(3, INPUT);
c) pinMode(3, OUTPUT); d) pinMode(3, ANALOG);

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a) Sound waves b) Ultrasonic waves

c) Radio waves d) Electromagnetic waves

34 Relate the purpose of the "Vin" pin on the Arduino board. [U] [1]

a) To interface with I2C devices b) To reset the microcontroller

c) To provide an external power source for the board d) To supply 3.3V to external components

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a) Pin 9 outputs 5V b) Pin 9 will read input voltage
c) Pin 9 will output 3.3V d) Pin 9 outputs 0V

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c) To reset the microcontroller d) To send digital data

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